

Kingdom of Science: chemistry.pdf

Soap Manufacturing

Alkaline metal hydroxides react with fatty acid triglycerides to yield soap. Required: Hardwood wood ash, water, animal fat, clay pot. Procedure: Leach ashes with hot water to extract lye liquid. Pour hot lye into melted fat while stirring continuously until trace stage. Safety: Lye is caustic.

Calcium Oxide Quicklime

Thermal decomposition of calcium carbonate at 900C releases CO₂ yielding CaO. Required: Clam/oyster shells, charcoal kiln, bellows. Procedure: Blast shells at 900C for 4 hours. Resulting white powder is Quicklime. Add water carefully for slaked lime. Safety: Heat and steam hazards.